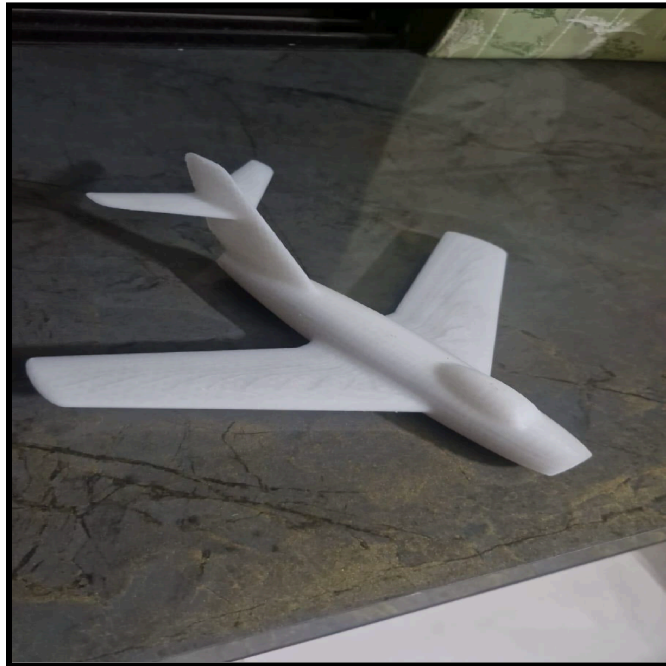


AERODYNAMICS 1

MEASUREMENT OF FORCES AND MOMENTS ON MiG 15 IN THE WIND TUNNEL



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Objective:

Evaluate the aerodynamic forces and moments exerted on a MiG-15 aircraft model in a wind tunnel to understand its performance characteristics. This involves measuring the six-component forces and moments: lift, drag, side force, rolling moment, pitching moment, and yawing moment. The goal is to use these precise measurements to refine the design and improve the MiG-15's aerodynamic efficiency and stability.

Testing procedure:

- 1. Attach a clamp to the model and set it up in the wind tunnel**
- 2. Start the wind tunnel**
- 3. Set the velocity to 5m/s**
- 4. Ensure 0 angle of attack initially**
- 5. Repeat with different angle of attacks**



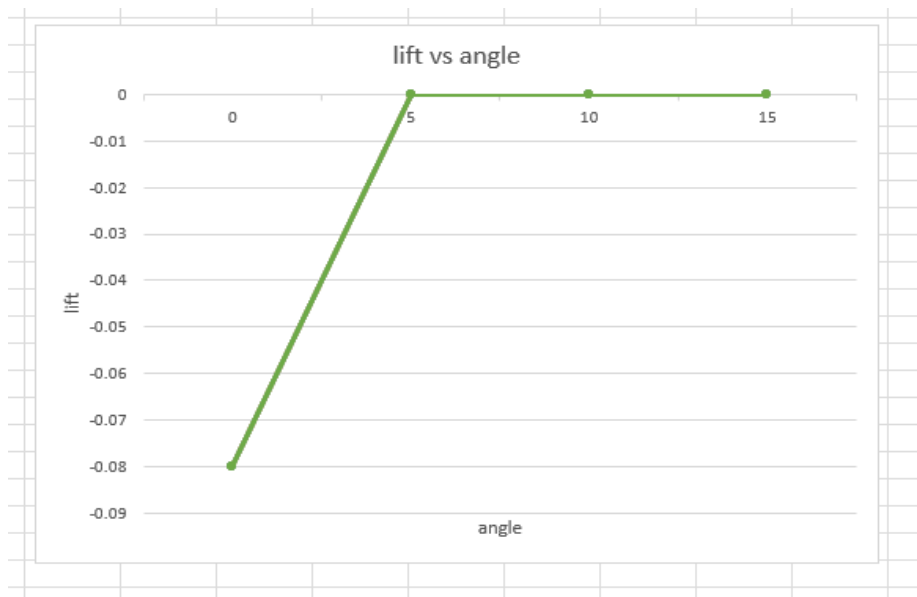
model set up in the wind tunnel



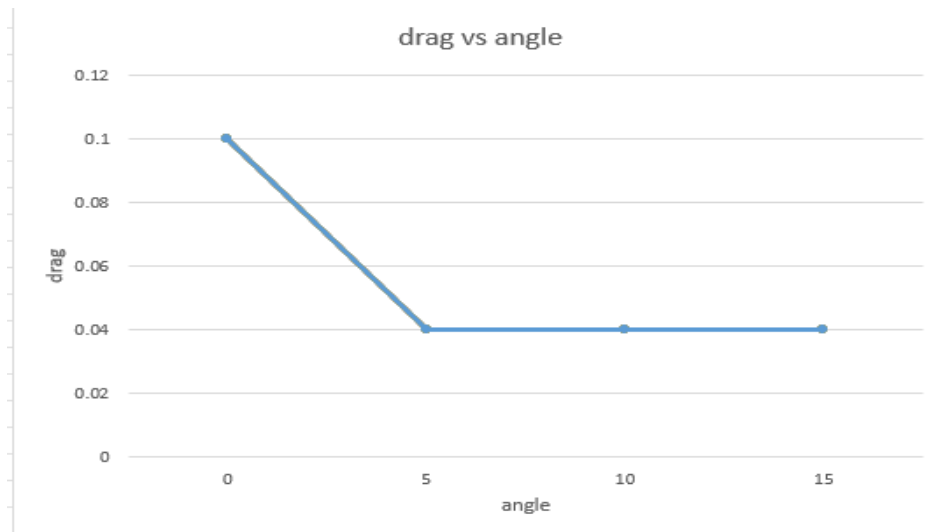
Observations At Velocity=5m/s:

Angle of attack	LIFT	DRAG	SIDE FORCE	PITCH	ROLL	YAW
0	-0.08	0.1	0	0.03	0	0
5	0	0.04	0	0.03	0	0
10	0	0.04	0	0.03	0	0
15	0	0.04	0	0.03	0	0

Results



Graph for angle of lift vs angle of attack



Graph for drag vs angle of attack

General characteristics of MiG 15

- Crew: 1 person
- Length: 10.1 meters (33.08 feet)
- Wingspan: 10.08 meters (33 feet)
- Height: 3.7 meters (12 feet)
- Wing Area: 20.6 square meters (221.75 square feet)
 - Empty Weight: 3,582 kilograms (7,892 pounds)
 - Gross Weight: 4,960 kilograms (10,935 pounds)
- Maximum Takeoff Weight: 6,105 kilograms (13,461 pounds)

Performance

- **Maximum Speed: 1,076 km/h (688 mph)**
 - **Cruise Speed: 840 km/h (520 mph)**
- **Range: 1,200 km (826 miles) without drop tanks, up to 2,220 km (1,379 miles) with 600L drop tank**
- **Service Ceiling: 15,500 meters (50,900 feet)**
- **Rate of Climb: 50 meters per second (9,842 feet per minute)**

Armament

- **Cannons: 1 x 37mm N-37D cannon with 40 rounds, 2 x 23mm NR-23 cannons with 80 rounds each**
- **Bombs: 2 x 100 kg (220 pounds) bombs or rocket pods on underwing hardpoints**

Engine

- **Type: Klimov VK-1 turbojet**
- **Thrust: 2,700 kg (5,952 pounds)**

Applications:

The MiG-15, a jet fighter developed by the Soviet Union, had several notable applications:

- **Air Superiority:**
- **Interceptor Missions**
- **Pilot Training**

Future developments:

The MiG-15, a legendary Soviet fighter jet, has undergone various developments and modifications over the years. While there aren't any direct future developments for the MiG-15, its legacy continues to influence modern aircraft design. Here are some potential areas of interest:

- **Experimental Variants:** The MiG-15 has been used as a testbed for various experimental projects, such as the **Samolet ST-10**, which tested ejection seats.

- **Upgrades and Modernization:** Some MiG-15 variants, like the MiG-15bis, have received upgrades, including improved engines, avionics, and armament.
- **Inspiration for Future Designs:** The MiG-15's design and capabilities have inspired future generations of fighter jets, including Russia's current projects like the MiG-41, a next-gen interceptor with advanced capabilities .

Acknowledgement

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